# Unit 8 Practice 1 — Transformations

Work through with your teacher · NC.8.G.2–4

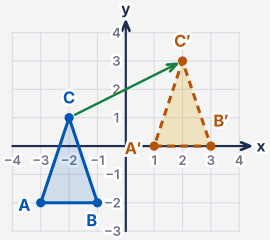
**ANSWER KEY**

```
Translation Rule: (x, y) → (x + h, y + k)  |  h > 0 → RIGHT  |  h < 0 → LEFT  |  k > 0 → UP  |  k < 0 → DOWN
```

**Remember:** Translations are rigid — the image is congruent to the pre-image (same size and shape, just shifted).

### Question 1 — Read the Translation

Score 3

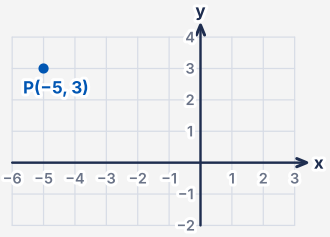



**blue** = pre-image · **gold dashed** = image

Triangle ABC (solid blue) was translated to A′B′C′ (dashed gold). Which rule was used?

- ○ (x − 4, y − 2)
- ● **(x + 4, y + 2)**
- ○ (x + 2, y + 4)
- ○ (x + 4, y − 2)

**Question 2** Score 3



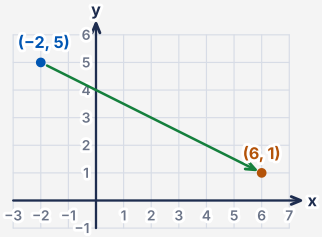

Rule: (x, y) → (x + 7, y − 4)

**blue** = pre-image · **gold dashed** = image

Point P(−5, 3) is translated using rule (x + 7, y − 4). Find P′.

- ● **P′(2, −1)**
- ○ P′(−12, 7)
- ○ P′(2, 7)
- ○ P′(−5, 7)

**Question 3** Score 4



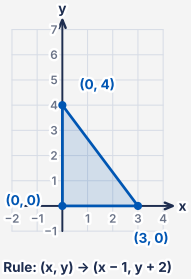

*count the moves, then answer*

**blue** = start · **gold** = end

A point moves from (−2, 5) to (6, 1). Write the translation rule.

- ● **(x + 8, y − 4)**
- ○ (x + 4, y − 4)
- ○ (x − 8, y + 4)
- ○ (x + 8, y + 4)

**Question 4** Score 4



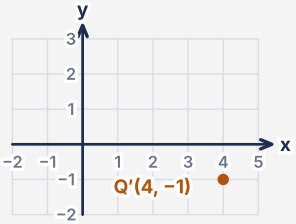

Rule: (x, y) → (x − 1, y + 2)

**blue** = pre-image · **gold dashed** = image

Translate triangle with vertices (0,0), (3,0), (0,4) by (x − 1, y + 2). What are all three image vertices?

- ● **(0,0) → (−1, 2), (3,0) → (2, 2), (0,4) → (−1, 6)**
- ○ (0,0) → (−1, −2), (3,0) → (2, −2), (0,4) → (−1, 2)
- ○ (0,0) → (1, 2), (3,0) → (4, 2), (0,4) → (1, 6)
- ○ (0,0) → (0, 2), (3,0) → (3, 2), (0,4) → (0, 6)

**Question 5** Score 5




Rule: (x, y) → (x + 5, y − 3)

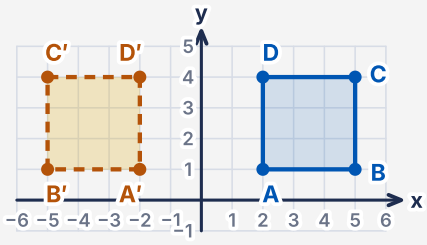
**blue** = pre-image · **gold dashed** = image

After translation (x + 5, y − 3), the image is at Q′(4, −1). Find the original point Q.

- ● **Q(−1, 2)**
- ○ Q(9, −4)
- ○ Q(4, −1)
- ○ Q(1, 2)

```
Over x-axis: (x, y) → (x, −y)  |  Over y-axis: (x, y) → (−x, y)
Memory trick: the axis you reflect OVER keeps its coordinate. Reflecting over x-axis → x stays, y flips.
```

**Question 1 — Read the Reflection** Score 3

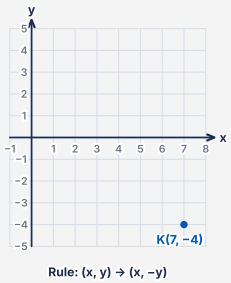



**blue** = pre-image · **gold dashed** = image

Rectangle ABCD was reflected. Over which line was it reflected?

- ○ x-axis
- ● **y-axis**
- ○ y = x
- ○ y = −x

**Question 2** Score 3

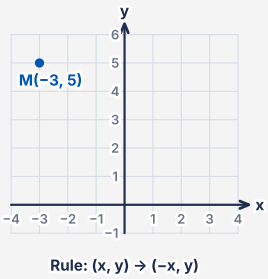



**blue** = pre-image · **gold dashed** = image

Point K(7, −4) is reflected over the x-axis. Find K′.

- ○ K′(−7, 4)
- ● **K′(7, 4)**
- ○ K′(−7, −4)
- ○ K′(4, 7)

**Question 3** Score 3

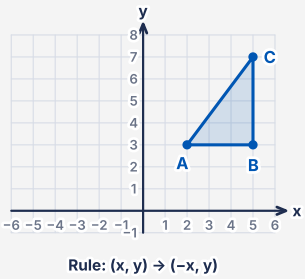



**blue** = pre-image · **gold dashed** = image

Point M(−3, 5) is reflected over the y-axis. Find M′.

- ● **M′(3, 5)**
- ○ M′(−3, −5)
- ○ M′(3, −5)
- ○ M′(5, −3)

**Question 4** Score 4

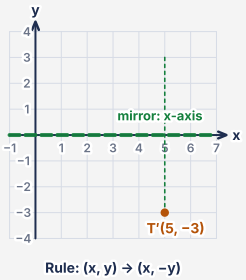



**blue** = pre-image · **gold dashed** = image

Triangle with vertices A(2,3), B(5,3), C(5,7) is reflected over the y-axis. Find A′, B′, C′.

- ● **A′(−2,3), B′(−5,3), C′(−5,7)**
- ○ A′(3,2), B′(3,5), C′(7,5)
- ○ A′(2,−3), B′(5,−3), C′(5,−7)
- ○ A′(−2,−3), B′(−5,−3), C′(−5,−7)

**Question 5** Score 5




Rule: (x, y) → (x, −y)

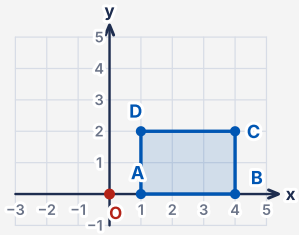
**blue** = pre-image · **gold dashed** = image

Point T(a, b) is reflected over the x-axis. The image is T′(5, −3). What are the values of a and b?

- ○ a = −5, b = 3
- ○ a = 5, b = −3
- ● **a = 5, b = 3**
- ○ a = 3, b = 5

```
90° CCW: (x,y)→(−y, x)  |  180°: (x,y)→(−x,−y)  |  270° CCW: (x,y)→(y,−x)
90° CW = 270° CCW  |  90° CCW = 270° CW
```

### Question 1 — 90° CCW Rotation — Score 3



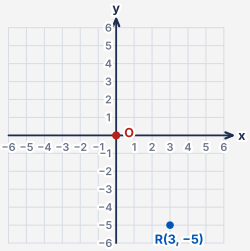

Rule: (x, y) → (−y, x)

**blue** = pre-image · **gold dashed** = image

Rectangle ABCD was rotated 90° CCW about the origin. What is the image of vertex B(4, 0)?

- ○ B′(4, 0)
- ○ B′(−4, 0)
- ● **B′(0, 4)**
- ○ B′(0, −4)

### Question 2 — Score 3



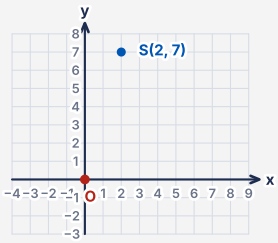

Rule: (x, y) → (−x, −y)

**blue** = pre-image · **gold dashed** = image

Point R(3, −5) is rotated 180° about the origin. Find R′.

- ○ R′(5, 3)
- ○ R′(3, 5)
- ● **R′(−3, 5)**
- ○ R′(5, −3)

### Question 3 — Score 4



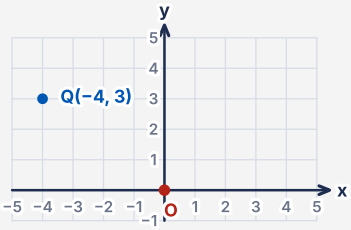

Rule: (x, y) → (y, −x)

**blue** = pre-image · **gold dashed** = image

Point S(2, 7) is rotated 270° CCW about the origin. Find S′.

- ○ S′(−2, −7)
- ● **S′(7, −2)**
- ○ S′(7, 2)
- ○ S′(−7, 2)

### Question 4 — Score 4



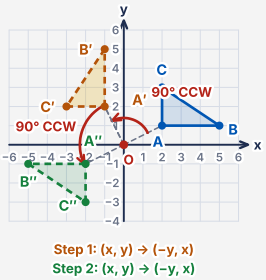

Rule: (x, y) → (y, −x)

**blue** = pre-image · **gold dashed** = image

Point Q(−4, 3) is rotated 90° CW (= 270° CCW) about the origin. Find Q′.

- ● **Q′(3, 4)**
- ○ Q′(4, 3)
- ○ Q′(−3, −4)
- ○ Q′(3, −4)

**Question 5** Score 5



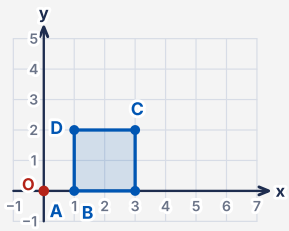

**blue** → **gold** (step 1) → **green** (step 2)

A triangle is rotated 90° CCW, then rotated another 90° CCW. What single rotation gives the same result?

- ○ 90° CCW
- ○ 270° CCW
- ● **180°**
- ○ 360°

```
Dilation: (x, y) → (kx, ky)  |  k > 1 = enlargement  |  0 < k < 1 = reduction  |  Similar figures (SAME shape, DIFFERENT size)
```

**Question 1 — Enlargement k = 2** Score 3



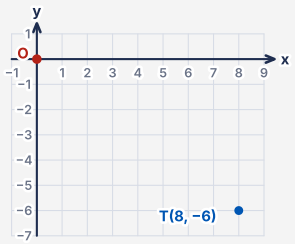

Rule: (x, y) → (2x, 2y)

**blue** = pre-image · **gold dashed** = image

Rectangle ABCD was dilated with k = 2. What are the coordinates of B′?

- ○ B′(3, 0)
- ○ B′(5, 2)
- ● **B′(6, 0)**
- ○ B′(4, 0)

**Question 2** Score 3



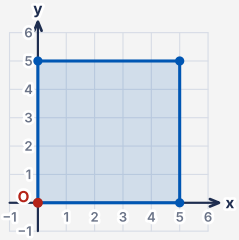

Rule: (x, y) → (½x, ½y)

**blue** = pre-image · **gold dashed** = image

Point T(8, −6) is dilated with scale factor k = ½. Find T′.

- ● **T′(4, −3)**
- ○ T′(4, 3)
- ○ T′(16, −12)
- ○ T′(−4, 3)

**Question 3** Score 4



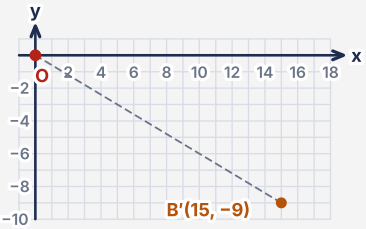

Rule: (x, y) → (⅘x, ⅘y)

**blue** = pre-image · **gold dashed** = image

A scale factor of k = ⅘ means the dilation is:

- ○ An enlargement — the image is larger
- ○ No change — k is close to 1
- ● **A reduction — the image is smaller**
- ○ An enlargement — fractions always enlarge

**Question 4** Score 5



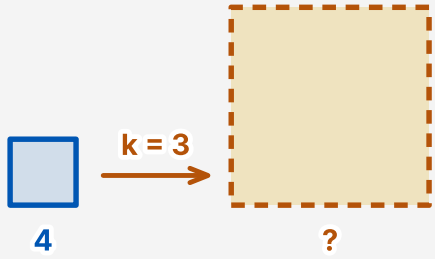

Rule: (x, y) → (3x, 3y)

**blue** = pre-image · **gold dashed** = image

After dilation, image B′ is at (15, −9). The scale factor was k = 3. Find the original B.

- ○ B(45, −27)
- ○ B(18, −12)
- ● **B(5, −3)**
- ○ B(3, −9)

**Question 5** Score 5



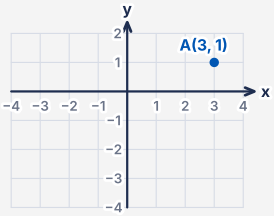

A square has side length 4 units. It is dilated with k = 3. What is the new side length? New perimeter? New area?

- ● **Side=12, Perimeter=48, Area=144**
- ○ Side=12, Perimeter=36, Area=48
- ○ Side=7, Perimeter=28, Area=49
- ○ Side=7, Perimeter=28, Area=144

```
Composite = two or more moves in a row. Do them ONE AT A TIME in the given order. Order matters!
```

**Congruence vs. Similarity:** All rigid (translation + reflection + rotation) → Congruent ≅ · Any dilation → Similar ~

**Question 1** Score 4



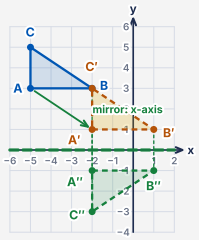

Step 1: (x, y) → (−x, y)
Step 2: (x, y) → (x + 2, y − 4)

**blue** → **gold** (step 1) → **green** (step 2)

Point A(3, 1): reflect over y-axis, then translate (x + 2, y − 4). Find the final coordinates.

- ● **(−1, −3)**
- ○ (−5, 3)
- ○ (5, −3)
- ○ (1, −3)

**Question 2** Score 4



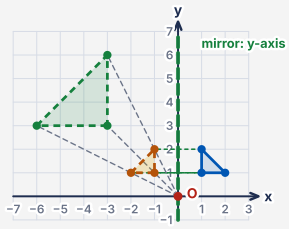

Step 1: (x, y) → (x + 3, y − 2)
Step 2: (x, y) → (x, −y)

**blue** → **gold** (step 1) → **green** (step 2)

A figure is translated (x+3, y−2), then reflected over the x-axis. Is the final image congruent or similar to the original?

- ○ Similar — translations change size
- ○ Similar — reflections change shape
- ● **Congruent — both are rigid transformations**
- ○ Congruent — but only if the translation is small

**Question 3** Score 4



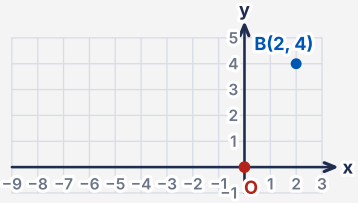

Step 1: (x, y) → (−x, y)
Step 2: (x, y) → (3x, 3y)

**blue** → **gold** (step 1) → **green** (step 2)

A figure is reflected over the y-axis, then dilated with k = 3. Congruent or similar?

- ○ Congruent — reflection preserves size
- ○ Congruent — if k is close to 1
- ● **Similar — the dilation changes size**
- ○ Neither — composite transformations always change shape

**Question 4** Score 5



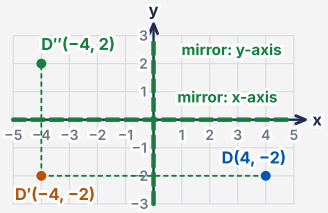

Step 1: (x, y) → (−y, x)
Step 2: (x, y) → (2x, 2y)

**blue** → **gold** (step 1) → **green** (step 2)

Point B(2, 4): rotate 90° CCW, then dilate k = 2. Find the final coordinates.

- ○ (−4, 4)
- ○ (4, 4)
- ● **(−8, 4)**
- ○ (−4, 8)

**Question 5** Score 5




Step 1: (x, y) → (−x, y)
Step 2: (x, y) → (x, −y)

**blue** → **gold** (step 1) → **green** (step 2)

Point D(4, −2): reflect over y-axis, then reflect over x-axis. What single transformation gives the same result?

- ○ Translation (x−8, y+4)
- ● **Rotation 180°**
- ○ Reflection over y = x
- ○ Dilation k = 2

## Study Tools

# Worked Examples

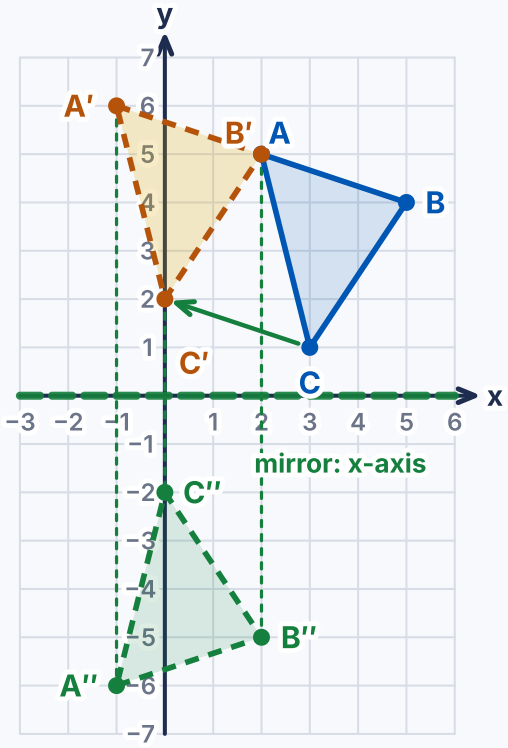
Click "Show Next Step" to reveal solutions one step at a time.

**Composite: translate △ABC — A(2,5), B(5,4), C(3,1) — by (x−3, y+1), then reflect over the x-axis**

**Step 1: Translate every vertex of △ABC by (x−3, y+1)**

`A(2,5)→A'(−1,6)  B(5,4)→B'(2,5)  C(3,1)→C'(0,2)`

Show Next Step →




**Step 1: slide 3 left, 1 up**
**Step 2: flip over x-axis**
A and B′ share the point (2, 5)

**△ABC** (solid blue) → translate → **△A′B′C′** (dashed gold) → reflect over x-axis → **△A″B″C″** (dashed green)

## Common Mistakes

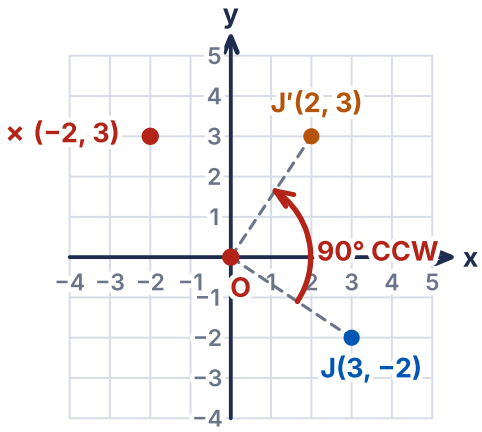
Can you spot what went wrong?

**Rotate △JKL 90° CCW — vertex J is at (3,−2)**

✗ **Student:** Student writes: (−2, 3)

✓ **Fix:** WRONG! Rule is (−y, x). y=−2, so −y = 2. Correct: (2, 3)

💡 *Be careful with negatives in rotation rules. −(−2) = +2.*



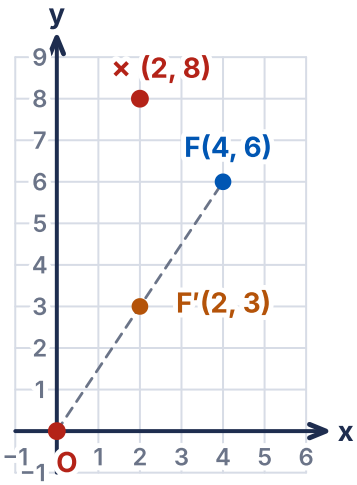

gold = correct · red × = student

**Dilate △FGH by k=½ — vertex F is at (4,6)**

✗ **Student:** Student writes: (2, 8)

✓ **Fix:** WRONG! ½ × 4 = 2 ✓ but ½ × 6 = 3, not 8. Correct: (2, 3)

💡 *Multiply BOTH coordinates by the scale factor.*




gold = correct · red × = student

## Key Tips for This Unit

✅ Translation: (x+a, y+b) — slide. CONGRUENT.

✅ Reflection: x-axis → negate y. y-axis → negate x. CONGRUENT.

✅ Rotation 90° CCW: (−y, x). 180°: (−x,−y). 270° CCW: (y,−x). CONGRUENT.

✅ Dilation: (kx, ky). SIMILAR (not congruent unless k=1).